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Volumetric Comparison of Hippocampal Regions in 44 Primate Species*

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With 6 Figures and 3 Tables

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Summary: Volumes of retrocommissural hippocampal regions (Subiculum, CA1, CA2, CA3, hilus region, and fascia dentata) were measured using serial sections of 44 species of prosimian and simian brains, including man. The volumes were compared allometrically with those of the least encephalized eutherian mammals, the madagascan tenrecs (Insectivora, Tenrecinae; 4 species).

The retrocommissural hippocampus is 2.9 times larger in prosimians, 2.4 times in non-human simians, and 4.1 times in man.

The different hippocampal regions do not enlarge uniformly to the same degree as the total hippocampus. The regions can be grouped into three categories, according to the degrees to which they enlarge in primates:

- a) Structures with the greatest degree of enlargement are the subiculum and the field CA1.
- b) An enlargement similar to that of the total retrocommissural hippocampus is seen for the hilus region.
- c) No or only a slight enlargement as compared to their sizes in tenrecs is observed for the hippocampal fields CA2, CA3, and the fascia dentata.

Key words: Cornu ammonis; Fascia dentata; Prosimians; Simians

Introduction

Since the classical descriptions of the hippocampus by RAMON y CAJAL (1903, 1911) many attempts have been made to subdivide this structure, using cyto- and myeloarchitectonic characteristics. The most widely accepted concept is that of LORENTE DE NÓ (1934), who distinguished the cornu ammonis fields CA1–CA4. Lesion experiments and tracing studies showed that the afferent and efferent connections differ among the various regions of the retrocommissural hippocampus (for review see CHRONISTER and WHITE, 1975; STEPHAN, 1975; WITTER, 1989). This finding emphasizes the functional aspects of LORENTE DE NÓ's (1934) parcellation.

Although the volumes of the hippocampus relative to the volumes of the brain as a whole decrease from an average of 9% in Insectivora to 4.5% in prosimians, 2.3% in non-human simians and 0.8% in man, allometric investigations (STEPHAN et al., 1988) have shown that the hippocampus is not a regressive, but a progressive structure in primate brain evolution. The relative decrease is not caused by a reduction in the size of the hippocampus, but by the enormous increase in the size of the neocortex.

STEPHAN and MANOLESCU (1980) measured the volumes of the pre-, supra- and retrocommissural hippocampus and found a decrease in the sizes of the precommissural hippocampus in prosimians and simians as compared to insectivores, whereas the supracommissural hippocampus was the region with the most pronounced increase of size in primates. The retrocommissural part of the hippocampus was also enlarged, but less than the supracommissural part. Furthermore, these authors measured the areal sizes of different regions of the retrocommissural hippocampus in a limited number of primates with the result, that CA1 showed the greatest enlargement in non-human primates and man.

Our report presents measurements of the volumes of retrohippocampal regions (CA1–CA3, fascia dentata, hilus region, subiculum) in a broad spectrum of primate species and evaluates which regions are most progressive in prosimians and simians as compared to Insectivora.

Material and Methods

Volumes of hippocampal regions were measured in four species of Insectivora and 44 species of primates (18 prosimians and 26 simians, including man). All brains were cut in serial sections and stained with various modifications of the Nissl stain. The sections belong to the Stephan collection, now located at the C. and O. Vogt Brain Research Institute, Düsseldorf.

* Dedicated to Dr. Heinz STEPHAN on the occasion of his 70th birthday.

The regions measured in the retrocommissural hippocampus are the subiculum, the cornu ammonis fields CA1, CA2 and CA3, the hilus region, and the fascia dentata.

For the delineation of the regions, cytoarchitectonic features were used, supplemented by adjacent serial sections stained for myelinated fibers. The criteria for the delineation are discussed in Results in connection with the description of the retrocommissural hippocampus in different species, and are depicted in Figs. 1 and 2. They are in accord with the criteria described in the classical works of RAMÓN Y CAJAL (1903, 1911) and LORENTE DE NÓ (1934).

Since the spacing between the sections was known, volumes were calculated from areal measurements of the regions delineated in camera lucida drawings of Nissl-stained sections, covering, at constant intervals, the entire extent of the retrocommissural hippocampus. These values were corrected for shrinkage due to histological procedures.

Allometric methods were applied for the evaluation of the data, i.e. the relationship between body weight and size of hippocampal structures was taken into account.

The allometric formula is:

$$y = b \cdot x^a$$

In its logarithmic form, it represents the formula for a straight line

$$\log y = \log b + a \cdot \log x$$

in which y is the size of the brain or brain structures, b the intercept of the allometric line with the abscissa, x is body weight, and a the slope of the line.

The slope of the allometric line for the hippocampus ($a = 0.57$) was taken from STEPHAN et al., 1991 (who calculated it by using a canonical discrimination analysis for a total of 56 families or subfamilies of Insectivora, Chiroptera, and Primates). In our study, their reference line was placed so as to run through the average body weight and hippocampal volumes (resp. its constituent parts) of the madagassian *Tenrecinae*, an insectivoran subfamily with the smallest brain sizes relative to body sizes found among mammals. All points on this line represent an index of 100. An index of e.g., 200 indicates that the relative size of brains or brain parts is twice as large as in the *Tenrecinae*.

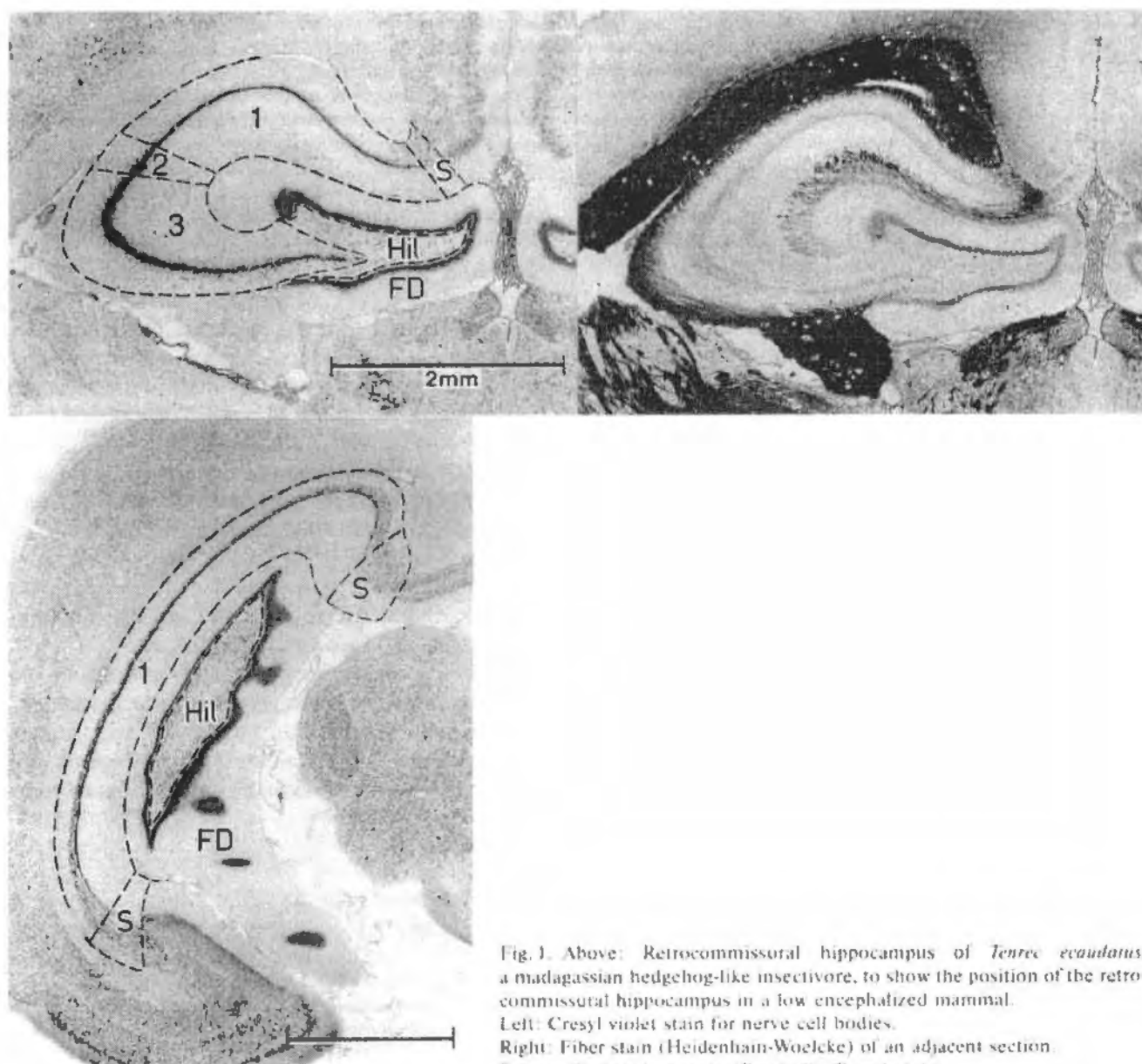


Fig. 1. Above: Retrocommissural hippocampus of *Tenrec ecaudatus*, a madagassian hedgehog-like insectivore, to show the position of the retrocommissural hippocampus in a low encephalized mammal. Left: Cresyl violet stain for nerve cell bodies. Right: Fiber stain (Heidenhain-Woelcke) of an adjacent section. Below: *Chetogaleus major* (Prosimii), Cresyl violet.

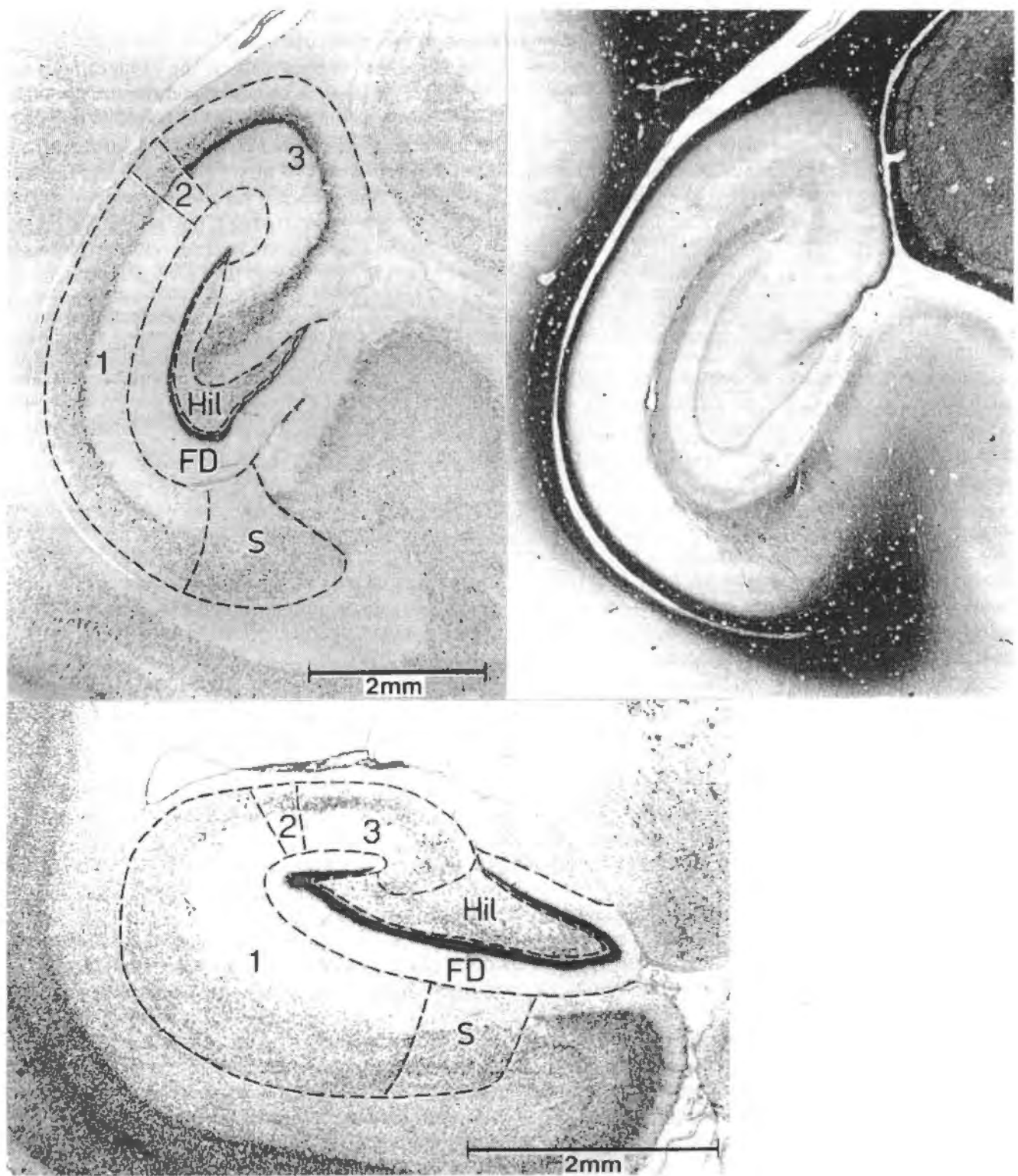


Fig. 2. Above: *Indri indri* (Prosimii)
 Left: Cresyl violet.
 Right: Fiber stain (Heidenhain-Woelcke) of an adjacent section.
 Below: *Miopithecus talapoin* (Simii), Cresyl violet.

Results

Coronal sections through the hippocampal region of representative species are presented in Figs. 1 and 2.

In the madagassian insectivore *Tenrec* (Fig. 1), the retrocommissural hippocampus is situated in a dorsal position, just beneath the corpus callosum. In the Nissl stain, CA 1 appears as a narrow, densely packed band of pyramidal cells. The subiculum is a small region, medial to CA 1, characterized by loosely packed cells; its border to the cingulate cortex is marked by the high density of cells in layer II of the retrosplenial cortex.

In insectivores, pyramidal cells in the hippocampal fields CA 2 and CA 3 are larger than those in CA 1, they stain more intensely and the layers of cells in CA 2 and CA 3 are thicker than those in CA 1. The border between CA 3 and CA 4 is very pronounced, since the dense cell layer of CA 3 ends abruptly and is replaced by the very loosely packed cells of CA 4. The CA 4 cells cannot be separated in Nissl-stained sections from the deep stratum multi-forme of the dentate gyrus and both fields were therefore measured together as the 'hilus region'. Few small blood vessels mark the border between the stratum moleculare of the fascia dentata and the lacunosum-moleculare layers of the cornu ammonis. This border is clearly seen in the fiber-stained adjacent section.

The tractus perforans, originating in the entorhinal region, passes through the subiculum from the deep white matter to a position close to the pial surface and then courses along the hippocampal sulcus to reach the fascia dentata.

The most prominent fiber bundle in the hippocampus is the fimbria/fornix complex. This complex is included in the measurements of the total hippocampus, but was excluded in this study of volumes of retrocommissural hippocampal fields.

The section from the hippocampus of the prosimian *Cheirogaleus* (Fig. 1) was taken from a rather posterior position in the hippocampus. CA 2 and CA 3 are not seen, since they have ended more rostrally. The subiculum can be seen in a dorsal position (as in *Tenrec*) between the CA 1 region and the retrosplenial cortex and again in a ventral position where it lies between CA 1 and the pre- and parasubicular regions. The cell layer of CA 1 is not as compact as in *Tenrec*. There is a narrow band of densely packed pyramidal cells on the surface of this layer facing the molecular layer, but cells tend to be more dispersed in the direction of the alveus, i.e. into the stratum oriens. Cells of the hilus region are totally surrounded by the granular cells of the fascia

dentata. Some small islands of cell bodies are also found in the molecular layer.

In Fig. 2 the ventral part of the hippocampus of the prosimian *Indri* is shown. The differentiation between the subiculum and the CA 1 field is not very clear, since the pyramidal cells of CA 1 form a broad band which is nearly as broad as the cell band of the subiculum. One criterion for the delineation is that the upper cell layer (toward the molecular layer) does not show as densely packed cells as CA 1 and the second criterion is that the very small cells in the deepest layer of the subiculum just above the white matter are very densely packed.

The border between CA 3 and the hilus region of the fascia dentata also tends to become blurred. The cell band of CA 3 becomes broader toward the fascia dentata, but the cells are still more densely packed and larger than those of the hilus region.

Due to the enormous enlargement of the neocortex in primates and the development of a temporal lobe, the hippocampus is rotated toward ventral and rostral, so that most of the hippocampus is situated on the medial side of the temporal lobe, in the hippocampal and parahippocampal gyri. This is seen in the Old world primate *Miopithecus* (Fig. 2). In this species, the cell layer of CA 1 is in some regions broader than that of the subiculum. The stratum oriens is not clearly demarcated in Nissl-stained sections, i.e. perikarya of pyramidal cells are found in close proximity to the alveus.

CA 3 is fairly small in this section, which was taken approximately from the middle of the rostro-caudal extent of the retrocommissural hippocampus, but it becomes much larger toward the caudal end. The hilus region is filled with densely packed cells (for comparison see e.g., *Tenrec*).

Volumes of the different retrocommissural regions of the hippocampus are given in Table 2. The volume of the total retrocommissural hippocampus is calculated as the sum of the volumes of these regions and given in the last column of Table 1. This part does not include the fibers of the fimbria/fornix complex, which extends far more rostral than the fields of the retrocommissural hippocampus proper. The volumes of these fibers are calculated together with the tiny pre- and supracommissural regions of the hippocampus. The volumes were not measured directly, but represent the differences between the sums of the retrocommissural hippocampal fields measured in the present study and the total hippocampus, as measured by STEPHAN et al. (1981, 1988), who used the same serial sections.

CA 3 is in absolute size the largest region of the retrocommissural hippocampus in the four tenrecine species. It is followed by CA 1 and the fascia dentata.

Table 1. Body weight (in g) and volumes (in mm³) of main regions of the hippocampus

Species*	Body weight	Hippocampus total	HP + HS + fibers	Hippocampus retrocomm.
<i>Tenrec ecaudatus</i>	852	147.12	22.59	124.53
<i>Setifer setosus</i>	237	96.57	11.54	85.03
<i>Hemicentetes semispinosus</i>	116	63.11	10.35	52.76
<i>Echinops telfairi</i>	87.5	46.66	4.65	42.01
<i>Microcebus murinus</i>	54	100.35	12.38	87.97
<i>Cheirogaleus major</i>	450	347.35	45.57	301.78
<i>Cheirogaleus medius</i>	177	174.36	23.75	150.61
<i>Lemur fulvus</i>	1400	752.10	96.46	655.64
<i>Varecia variegata</i>	3000	1403.65	226.70	1176.95
<i>Lepilemur ruficaudatus</i>	915	392.28	72.73	319.55
<i>Avahi laniger laniger</i>	1285	526.25	67.70	458.55
<i>Avahi l. occidentalis</i>	860	475.58	56.15	419.43
<i>Propithecus verreauxi</i>	3480	1042.96	160.88	882.08
<i>Indri indri</i>	6250	1529.38	358.58	1170.80
<i>Daubentonia madagascariensis</i>	2800	1776.10	328.20	1447.90
<i>Loris tardigradus</i>	322	191.14	24.56	166.58
<i>Nycticebus coucang</i>	800	565.77	67.06	498.71
<i>Perodicticus potto</i>	1150	607.01	78.69	528.32
<i>Galago senegalensis</i>	186	260.78	15.47	245.31
<i>Otolemur crassicaudatus</i>	850	460.88	73.43	387.45
<i>Galagoides demidoff</i>	81	152.24	21.76	130.48
<i>Tarsius spec.</i>	125	152.66	22.78	129.88
<i>Callithrix jacchus</i>	280	221.12	40.63	180.49
<i>Cebuella pygmaea</i>	120	122.34	22.84	99.50
<i>Saguinus midas tamarin</i>	340	280.26	53.55	226.71
<i>Saguinus oedipus</i>	380	262.09	53.10	208.99
<i>Callimico goeldi</i>	480	280.90	49.34	231.56
<i>Cebus albifrons</i>	3100	890.40	313.85	576.55
<i>Aotus trivirgatus</i>	830	538.80	133.49	405.31
<i>Callicebus moloch</i>	900	588.29	101.92	486.37
<i>Saimiri sciureus</i>	660	351.44	103.06	248.38
<i>Pithecia monachus</i>	1500	834.26	188.08	646.18
<i>Alouatta seniculus</i>	6400	1319.79	315.03	1004.76
<i>Ateles geoffroyi</i>	8000	1366.01	370.37	995.64
<i>Lagothrix lagothricha</i>	5200	1585.69	458.90	1126.79
<i>Cercopithecus albigena</i>	7900	1464.21	426.37	1037.84
<i>Papio anubis</i>	25000	3397.90	1318.88	2079.02
<i>Cercopithecus ascanius</i>	3400	1189.00	249.60	939.40
<i>Cercopithecus mitis</i>	6300	1366.39	379.53	986.86
<i>Miopithecus talapoin</i>	1200	704.87	190.75	514.12
<i>Erythrocebus patas</i>	7800	1590.53	504.92	1085.61
<i>Colobus badius</i>	7000	1670.84	221.64	1449.20
<i>Pygathrix nemaeus</i>	7500	2295.41	445.75	1849.66
<i>Nasalis larvatus</i>	14000	1965.93	518.83	1447.10
<i>Hylobates lar</i>	5700	2672.56	444.65	2227.91
<i>Pan troglodytes</i>	46000	3778.69	1015.23	2763.46
<i>Gorilla gorilla</i>	105000	4780.48	1475.05	3305.43
<i>Homo sapiens</i>	65000	10287.38	2853.78	7433.60

* Body weights and volumes of total hippocampus for Insectivora taken from STEPHAN et al., 1991, and for Primates from STEPHAN et al., 1981.

HP = Hippocampus praecommissuralis

HS = Hippocampus supracommissuralis

Table 2. Volumes of retrohippocampal regions (in mm³)

Species	Subiculum	CA 1	CA 2	CA 3	Hilus	Fascia dentata
<i>Tenrec ecaudatus</i>	7.50	32.11	6.49	39.70	8.00	30.73
<i>Setifer setosus</i>	5.83	21.24	5.67	25.79	6.63	19.87
<i>Hemicentetes semispinosus</i>	1.34	12.10	2.81	19.03	3.32	14.16
<i>Echinops telfairi</i>	2.00	12.21	3.11	13.67	2.30	8.72
<i>Microcebus murinus</i>	5.99	30.45	2.33	23.82	4.48	20.90
<i>Cheirogaleus major</i>	18.90	107.42	8.32	92.66	13.22	61.26
<i>Cheirogaleus medius</i>	9.97	54.40	4.72	44.89	6.19	30.44
<i>Lemur fulvus</i>	62.23	258.65	19.52	171.72	37.46	106.06
<i>Varecia variegata</i>	80.15	410.99	26.84	304.87	118.25	235.85
<i>Lepilemur ruficaudatus</i>	31.58	128.48	7.28	78.29	12.62	61.30
<i>Avahi laniger laniger</i>	48.13	178.89	15.29	112.49	24.93	78.82
<i>Avahi l. occidentalis</i>	41.61	175.41	9.13	90.39	30.86	72.03
<i>Propithecus verreauxi</i>	72.19	361.75	18.70	217.12	70.93	141.39
<i>Indri indri</i>	98.11	477.66	35.77	336.04	57.87	165.35
<i>Daubentonia madagascariensis</i>	100.01	577.11	35.54	316.55	94.14	324.55
<i>Loris tardigradus</i>	12.96	57.60	4.87	49.81	8.00	33.34
<i>Nycticebus coucang</i>	52.63	178.14	17.34	135.76	23.62	91.22
<i>Perodicticus potto</i>	48.35	197.36	16.30	139.69	26.62	100.00
<i>Galago senegalensis</i>	23.14	99.58	5.96	52.43	13.48	50.72
<i>Otolemur crassicaudatus</i>	42.22	164.58	9.60	78.22	28.14	64.69
<i>Galagoides demidoff</i>	13.40	53.70	3.08	28.64	7.00	24.66
<i>Tarsius spec.</i>	10.76	51.39	7.03	32.09	5.87	22.74
<i>Callithrix jacchus</i>	11.48	73.07	7.49	42.82	12.29	33.34
<i>Cebuella pygmaea</i>	7.45	38.83	4.11	26.95	4.49	17.67
<i>Saguinus midas tamarin</i>	15.85	78.04	10.95	68.92	13.97	38.98
<i>Saguinus oedipus</i>	23.45	85.42	9.66	49.77	9.14	31.55
<i>Callimico goeldi</i>	17.57	83.53	7.84	60.81	12.08	49.73
<i>Cebus albifrons</i>	43.85	210.19	30.29	146.01	35.87	110.34
<i>Aotus trivirgatus</i>	27.43	150.16	20.83	97.14	33.52	76.23
<i>Callicebus moloch</i>	42.71	174.88	22.90	129.11	37.33	79.44
<i>Saimiri sciureus</i>	17.58	79.82	13.70	66.43	21.07	49.78
<i>Pithecia monachus</i>	41.63	259.52	31.78	154.43	56.38	102.44
<i>Alouatta seniculus</i>	117.67	362.93	56.59	236.05	67.88	163.64
<i>Ateles geoffroyi</i>	72.63	388.30	51.79	263.35	69.29	150.28
<i>Lagothrix lagothricha</i>	78.34	400.11	49.57	332.56	73.93	192.28
<i>Cercocebus albigena</i>	73.47	395.87	38.15	304.20	69.90	156.25
<i>Papio anubis</i>	156.72	800.97	66.37	564.35	178.36	312.25
<i>Cercopithecus uscanius</i>	77.57	362.05	23.02	240.60	85.70	150.46
<i>Cercopithecus mitis</i>	77.48	378.84	30.44	288.47	86.05	125.58
<i>Miopithecus talapoin</i>	41.31	183.40	15.31	134.60	51.91	87.59
<i>Erythrocebus patas</i>	76.65	345.54	39.46	327.74	101.09	195.13
<i>Colobus badius</i>	129.20	615.79	38.44	323.72	156.77	185.28
<i>Pygathrix nemaeus</i>	117.14	786.60	44.48	420.73	227.64	253.07
<i>Nasalis larvatus</i>	113.27	546.36	69.39	370.10	142.29	205.69
<i>Hylobates lar</i>	139.39	866.51	76.73	547.00	226.25	372.03
<i>Pan troglodytes</i>	266.58	1166.33	85.70	622.88	235.43	386.54
<i>Gorilla gorilla</i>	191.70	1405.57	130.65	951.51	210.02	415.98
<i>Homo sapiens</i>	821.76	2793.05	229.52	2357.63	397.17	834.47

Table 3. Allometric size indices of encephalization, hippocampus, and hippocampal regions.

Species	ENC	HIP	fiber	HR	SUB	CA 1	CA 2	CA 3	Hilus	FD
<i>Leontideus rosalia</i>	86	82	94	81	97	81	68	78	79	83
<i>Sciurus setosus</i>	117	112	100	114	156	111	122	106	136	112
<i>Hemicentetes semispinosus</i>	104	110	135	106	54	95	91	117	102	119
<i>Echinops telfairi</i>	93	96	71	99	94	113	118	99	83	86
Mean, leontes (n = 4)	100	100	100	100	100	100	100	100	100	100
<i>Microcebus murinus</i>	366	271	248	274	372	370	117	227	213	273
<i>Cheirogaleus major</i>	345	280	274	281	350	390	125	263	188	239
<i>Cheirogaleus medius</i>	295	239	243	239	314	336	120	217	150	202
<i>Lemur fulvus</i>	558	318	303	319	604	491	153	256	279	216
<i>Varecia variegata</i>	456	384	462	371	503	506	136	294	570	312
<i>Lepilemur ruficaudatus</i>	241	211	292	198	390	311	73	148	120	159
<i>Avahi laniger laniger</i>	266	233	224	235	490	357	126	176	195	169
<i>Avahi l. occidentalis</i>	320	265	233	270	533	440	95	178	303	194
<i>Propithecus verreauxi</i>	351	262	301	256	417	409	87	192	314	172
<i>Indri indri</i>	342	275	481	243	406	387	120	213	184	144
<i>Daubentonius madagascariensis</i>	685	505	695	475	653	739	188	317	472	446
<i>Loris tardigradus</i>	417	187	179	188	290	253	88	171	138	157
<i>Nycticebus coucang</i>	433	329	290	334	702	466	187	278	242	256
<i>Perodicticus potto</i>	382	287	277	288	525	419	143	233	222	228
<i>Galago senegalensis</i>	436	348	154	378	709	598	148	247	317	327
<i>Otolemur crassicaudatus</i>	343	259	307	251	544	416	100	155	278	175
<i>Galagoides demidoff</i>	531	326	347	323	660	518	123	216	265	255
<i>Tarsius spec.</i>	425	256	284	251	414	387	219	189	173	184
Mean, prosimians (n = 18)	400	291	311	287	493	433	130	221	257	228
<i>Callithrix jacchus</i>	527	234	320	220	279	347	147	160	229	170
<i>Cebuella pygmaea</i>	503	210	291	197	293	299	131	163	135	146
<i>Saguinus midas tamarin</i>	628	265	377	247	344	332	193	230	233	178
<i>Saguinus oedipus</i>	567	233	351	214	478	341	159	156	143	135
<i>Callimico goeldi</i>	534	218	286	208	314	292	113	167	165	187
<i>Cebus albifrons</i>	1007	239	628	179	270	254	151	138	170	143
<i>Aotus trivirgatus</i>	579	307	566	266	358	384	220	195	336	210
<i>Callicebus moloch</i>	609	320	412	305	533	427	231	247	357	209
<i>Saimiri sciureus</i>	945	228	498	186	262	233	165	152	241	156
<i>Pithecia monachus</i>	801	339	569	303	388	474	240	221	403	201
<i>Alouatta seniculus</i>	457	234	417	206	480	290	187	148	212	140
<i>Ateles geoffroyi</i>	819	214	431	180	261	273	150	145	191	114
<i>Lagothrix lagothricha</i>	1018	317	683	260	360	360	184	234	260	186
<i>Cercopithecus albigena</i>	795	231	500	189	266	280	112	169	194	119
<i>Papio anubis</i>	719	278	802	196	294	294	101	162	257	123
<i>Cercopithecus ascanius</i>	894	303	473	276	454	415	109	216	385	185
<i>Cercopithecus mitis</i>	666	245	507	204	319	305	101	182	272	109
<i>Miopithecus talapoin</i>	1061	325	655	273	437	380	131	219	422	195
<i>Erythrocebus patas</i>	833	252	597	199	279	247	116	183	283	150
<i>Colobus badius</i>	646	282	279	282	501	467	121	193	466	151
<i>Pygathrix nemaeus</i>	609	372	539	346	436	574	134	241	651	198
<i>Nasalis larvatus</i>	509	223	439	190	296	279	147	148	285	113
<i>Hylobates lar</i>	968	507	628	488	607	739	270	366	756	341
<i>Pan troglodytes</i>	968	218	436	184	353	303	92	127	239	108
<i>Gorilla gorilla</i>	693	172	396	137	159	228	88	121	133	72
Mean, non-human simians (n = 25)	734	271	483	237	361	353	152	187	297	162
<i>Homo sapiens</i>	2531	487	1007	406	894	595	202	394	332	191

Abbreviations:

CA 1-3 = fields of cornu ammonis

ENC = Encephalization

FD = Fascia dentata

fiber = fimbria/fornix complex + pre- and supracommissural hippocampus

HIP = Hippocampus, total

HR = retrocommissural hippocampus

SUB = Subiculum

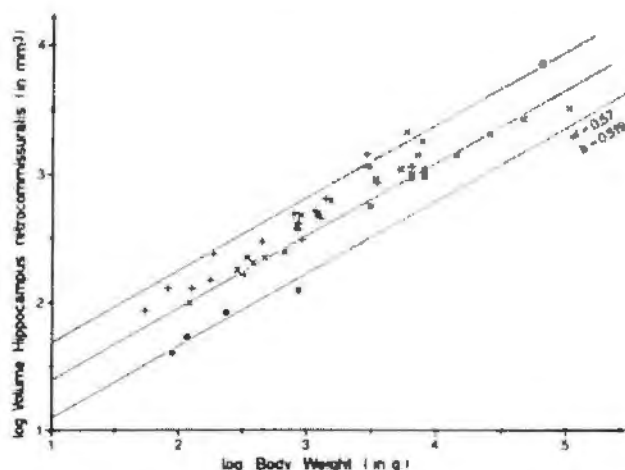


Fig. 3. Log-log plot of retrocommissural hippocampus volume against body weight. The allometric reference line ($a = 0.57$, $b = 0.519$) runs through the four species of Tenrecinae, the lowest encephalized insectivores. The parallel lines indicate a two- resp. fourfold enlargement of the hippocampus.

● = Tenrecinae + = Prosimians
 × = Non-human simians * = Homo

In all primate species however, we found that the CA 1 field is the largest in absolute size, with CA 3 as the second largest and the fascia dentata in the third position. Only in *Daubentonia* were the positions of CA 3 and fascia dentata reversed, with the volume of the fascia dentata slightly larger than that of CA 3. The sizes of hippocampal regions are obviously not uniformly enlarged in primates as compared to insectivores. The extent to which the development of hippocampal regions in primates is inhomogeneous, can be shown using allometric methods.

To illustrate the allometric approach, two log-log plots of hippocampal regions are presented. In Fig. 3 the volume of the retrocommissural hippocampus is plotted against the body weight. The reference line runs through the mean of the four Tenrecinae species (their average size index is 100). There is a clear enlargement in the size of this hippocampal part in primates, but the prosimian and simian values overlap.

The values for the field CA 1 of prosimians and simians also overlap (Fig. 4). Furthermore, the distance of primate values from the Tenrecinae reference line is in general greater, indicating that the progression of CA 1 in primates is more pronounced than that of the retrocommissural hippocampus as a whole.

The distances from the reference line are expressed as size indices and scaled in Figs. 5 and 6. In this mode of presentation, the average enlargement factors for prosimians, simians and man and the scatter of size indices within these groups can be

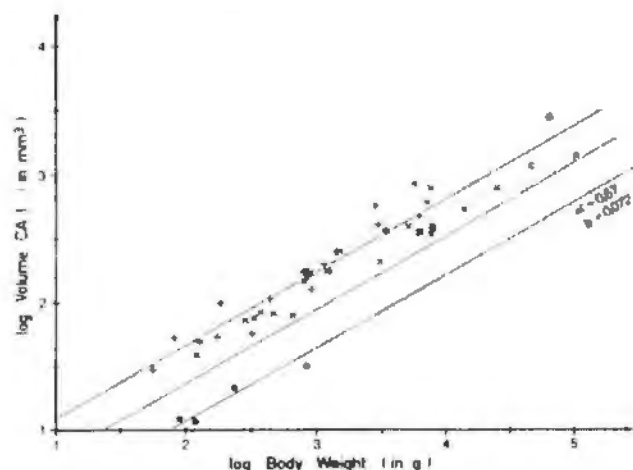


Fig. 4. Log-log plot of CA 1 volume against body weight. Most primate species show a greater progression of this field than for the total hippocampus. Symbols as in Fig. 3.

directly compared for all hippocampal regions measured.

The size indices of the total hippocampus range from 187 (*Loris*) to 505 (*Daubentonia*) in prosimians and from 172 (*Gorilla*) to 507 (*Hylobates*) in simians, which means that there is a complete overlap of size indices for these two groups of primates and no further increase in the size of the simian hippocampus above the size reached by prosimians. The average index is 291 for prosimians and 271 for simians, which indicates that the relative size of the total hippocampus in primates is 2.8 times as large as that of the least encephalized mammals, the madagascan Tenrecinae.

Nearly the same ranges and averages as for the total hippocampus were observed for its retrocommissural part. This is not surprising, since it is the largest part, comprising more than 90% of the total hippocampus.

Within the retrocommissural hippocampus, the regions differ as to the development of their allometric sizes.

Subiculum: In tenrecine species, the size indices for the subiculum range from 54 to 156, which is a rather large deviation from the average of 100. The indices are much higher for prosimians (range: 290–709, average: 493), for non-human primates (range: 159–607, average 361) and for man (894). The overlap of size indices for prosimians and simians is almost complete.

CA 1: There is a large gap between the size indices for tenrecines and primates; CA 1 is clearly enlarged

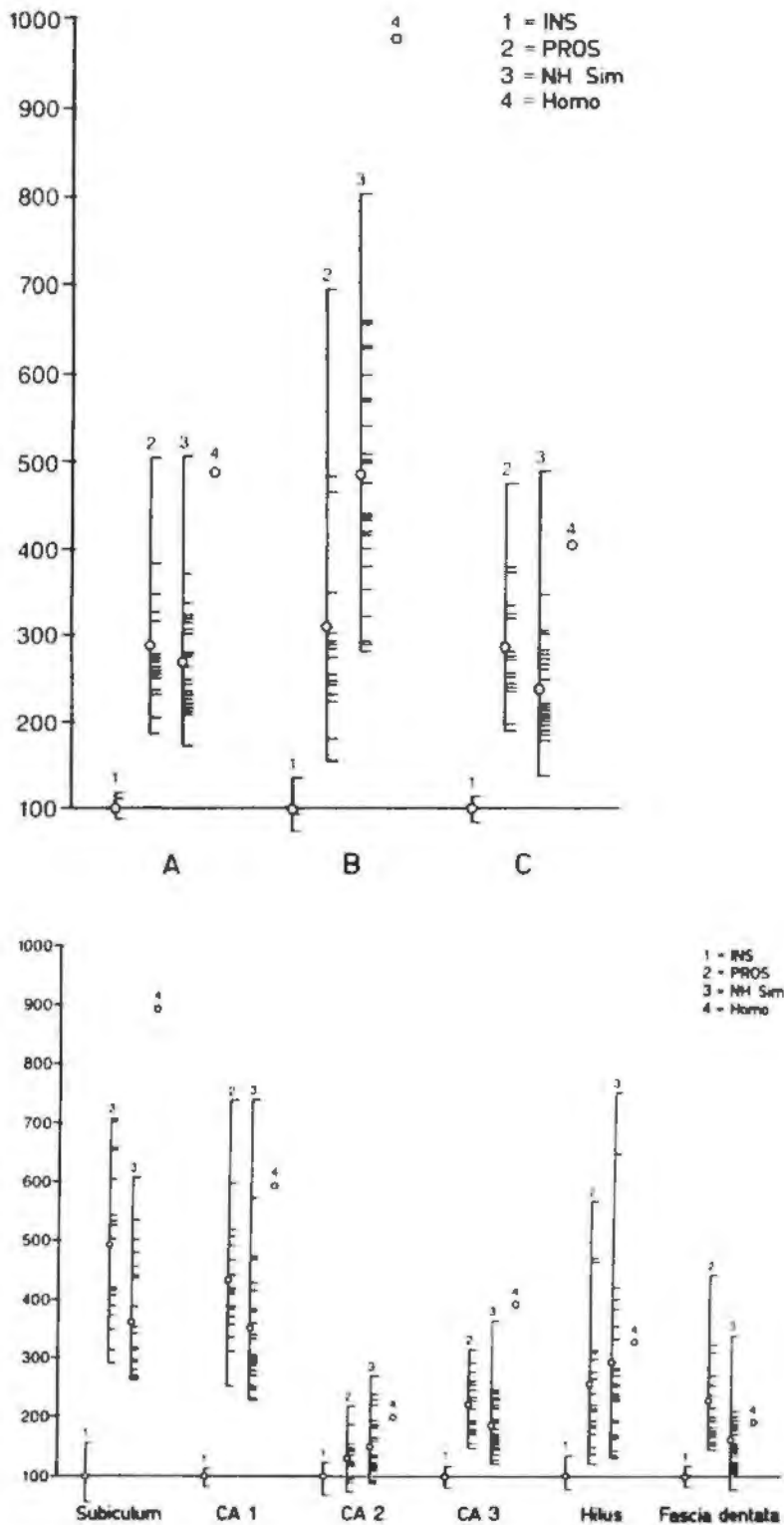


Fig. 5. Allometric size indices (a numerical expression for the deviation of the species from the tenrecine reference line) for A: Total hippocampus (= pre-, supra-, and retrocommissural).

B: Component made up of the tiny pre- and supracommissural hippocampus and the fibers of the fimbria/fornix complex.

C: Retrocommissural hippocampus (large fiber tracts excluded).

In the vertical columns of Insectivores (1), prosimians (2), non-human simians (3), and man (4), the single species are indicated by a small horizontal bar, the average index of a group by a circle.

Fig. 6. Allometric size indices for the different regions of the retrocommissural hippocampus. Symbols and abbreviations as in Fig. 5.

in primates. The range of size indices covered by prosimians (253–739) and non-human simians (228–739) is nearly identical; the size index for man (595) does not surpass the range of the other primates.

CA2: This region shows the slightest progression in primates as compared to Tenrecinae. The ranges of both prosimians and simians overlap with those of insectivores. The average indices are 130 for prosimians and 152 for simians.

CA3: It shows, similar to CA2, only a slight progression in primates (prosimians: range 148–317; simians: range 121–366; man 394), but in contrast to CA2 there is no overlap with the size indices for tenrecines.

Hilus: This region is more progressive than CA2 and CA3 in most primate species. It exhibits the largest range of size indices in non-human simians observed among all retrohippocampal regions. Some species (as e.g., *Lepilemur*, *Cebuella*, and *Gorilla*) have indices in the range of tenrecines, but others (*Hylobates*, *Pygathrix*, *Varecia*) show an enlargement of more than 5 times.

Fascia dentata: In prosimians, the indices of this region are higher than those of tenrecines (range: 144–446; average: 228), whereas among non-human simians (range: 72–341; average: 162) some species tend to have lower indices than prosimians, reaching as low as the range covered by tenrecs.

In conclusion, the retrocommissural regions of the primate hippocampus can be arranged in three categories: The first group of structures shows a clear enlargement as compared to those of tenrecines and their indices are also higher than those of the total retrocommissural hippocampus. The second group is intermediate, which means that the structures occupy on the average a position similar to that of the retrocommissural hippocampus. The third group of structures is not, or only slightly, progressive as compared to those of insectivores, but their indices lie below those for the retrocommissural hippocampus.

The retrocommissural hippocampus of primates shows an average enlargement of 2.6 in comparison to that of tenrecs.

Even more progressive than the total retrocommissural hippocampus are two regions, the subiculum and field CA1. They belong to the first category with a clear enlargement.

The hilus region belongs to the second category. Its average indices for prosimians and simians are similar to those of the retrocommissural hippocampus, although the range of indices is much broader.

The hippocampal fields CA2, CA3, and the fascia dentata fall into the third category with no or only slight enlargements compared to those of tenrecs.

The average size indices of these fields are below those of the retrocommissural hippocampus.

One further subdivision of the hippocampus must be mentioned briefly: When comparing the size indices of the total hippocampus with those of its retrocommissural part, it can be seen that the averages as well as minimal and maximal values are slightly lower for the retrocommissural part. This indicates that the supra- and precommissural parts of the total hippocampus and the fimbria/fornix complex are more progressive than its retrocommissural part. It can be seen in Fig. 5 that the fimbria/fornix is very progressive, reaching a maximum size index of 1000 in man (e.g. 10 times larger than in tenrecs). Volume measurements of the pre- and supracommissural hippocampus in some primate species may give a hint, which of the parts in this "mixed" component contributes most to this progression of indices.

Discussion

Comparative metric investigations on subdivisions of the retrocommissural hippocampus in insectivores and primates were performed by STEPHAN and MANOLESCU (1980). They used a broad range of Insectivora species, but only a few primates (3 prosimians, 3 non human simians, and man) were included to show trends in the evolution of sizes of hippocampal regions in primates. They did not measure volumes, arguing that in Nissl-stained sections there are no distinct characteristics, especially in the broad strata moleculare and oriens of the cornu ammonis fields, which would allow a precise delineation between these regions. This is true for the majority of the insectivoran species they investigated, but is not critical in primates, which always show a clear hippocampal sulcus, separating the molecular layers of the fascia dentata from those of the CA fields. Furthermore, fiber stained sections can provide additional information for a delineation, since they show differences in staining intensities of fine fibers in the radiatum, lacunosum and molecular layers between the CA fields (see Figs. 1 and 2).

STEPHAN and MANOLESCU (1980) measured the lengths of the pyramidal and granule cell layers of the different regions and calculated areas by multiplying them with the distance between the sections. This results in correct values only when the layers are cut perpendicularly. If the layers are cut obliquely, the values are always too small. These authors therefore introduced a correction factor for oblique cutting, deduced from three brains, and

applied it to all their measurements. One further restriction is that structures which do not have a compact cell band cannot be measured with this method, as, for instance, the scattered cells of CA 4 in the hilus of the dentate gyrus.

Our volume measurements do not have this limitation, nor did we need to introduce correction factors for oblique cutting. The difficulties in delineating CA fields can, in most cases, be solved when using corresponding fiber stains and are generally smaller in primates than in insectivores. In cases where no border could be found in the strata radiatum, lacunosum and moleculare, the line was drawn radially from the pyramidal cell band to the pial surface.

STEPHAN and MANOLESCU (1980) found that CA 1 is the most progressive of the hippocampal regions and that in man the subiculum is also very large. These structures are followed in relative size by the fascia dentata and finally, CA 2/3 show the least progression in man as compared to Insectivora.

The absolute height of indices cannot directly be compared between the two investigations, since the reference group is different. STEPHAN and MANOLESCU used ten Insectivora species; in the present study, four species of Tenrecinae (as representatives of the least encephalized insectivoran group) were chosen.

The trends in the enlargement of hippocampal regions, however, can be fully confirmed. The following conclusions can be drawn from volume measurements in a broad spectrum of primates:

- a) CA 1 is, together with the subiculum, one of the most progressive regions of the hippocampus. The index for man is not as exceptionally high as suggested by the STEPHAN and MANOLESCU (1980) data; some other simian and even prosimian species show indices as high as man.
- b) Indices for the subiculum are in general not smaller, in some species they are even higher than those of CA 1. This was not seen so clearly in areal measurements. The reason for this difference between the two methods may lie in the fact that the subiculum tends to have a broader molecular layer in many sections than the fields of the cornu ammonis. The size of the molecular layer cannot be determined by STEPHAN's and MANOLESCU's (1980) technique.
- c) Although some maximal indices are larger for the fascia dentata than for CA 3, there are also species with indices as low as those of tenrecs. Judged from volume measurements, the fascia dentata is not more progressive than CA 3; both, fascia dentata and CA 3 fall into the same category of regions with a slight enlargement.

d) Due to their technique, STEPHAN and MANOLESCU (1980) were not able to measure the hilus region. It shows a very large range of indices for the various primates species, but generally maintains a medium position, i.e., the hilus region tends to be more progressive than CA 2, CA 3, and the fascia dentata, but less so than the subiculum or CA 1. In fact, the average indices of the hilus region are similar to those of the retrocommissural hippocampus as a whole.

e) CA 2, a region not measured separately by STEPHAN and MANOLESCU, shows the least progression of all retrocommissural fields. Many indices for prosimian as well as simian species do not exceed the range of indices covered by insectivores. From his examinations of histological sections, ANGEVINE (1975) states that the area dentata has enlarged especially in primates and man. The undulations of the granular cell layer in some primate species may, of course, give the impression of an extraordinary enlargement of this region, whereas such undulations are not observed in the pyramidal layers of CA fields. The volumetric comparisons show that the fascia dentata is indeed only slightly enlarged as compared to that of tenrecs, much less than the subiculum and the field CA 1.

More progressive than the retrocommissural hippocampus is the component consisting of the pre- and supracommissural hippocampus and the hippocampal fibers (mainly the alveus and the fimbria/fornix complex).

STEPHAN and MANOLESCU (1980) measured the volumes of these main subdivisions. The precommissural hippocampus is a tiny structure, occupying between 0.6% (man) and 5.4% (insectivores) of the total hippocampus. In prosimians it is 2% and in non-human simians 1.3%.

The supracommissural hippocampus increased its percentage values (relative to the total hippocampus) from 1.1% in insectivores and prosimians to approximately 3% in simians.

The allometric indices showed a decrease in the size of the precommissural hippocampus in prosimians; in this group it was 3/4 of its size in insectivores and in simians it was only half as large.

On the other hand, the supracommissural hippocampus increased 2.2 fold in prosimians and even more in non-human simians and in man.

Subtracting the volumes of the pre- and supracommissural hippocampus reveals that the fibers of the fimbria/fornix complex contribute most to the volumes of this mixed (HP, HS + fibers) component in primates. Therefore, the high progression of this complex seems to be due mainly to the volumetric increase in fibers tracts.

This progression may be seen in connection with the development of the temporal lobe in primates and the displacement of the retrocommissural hippocampus from a position close to the corpus callosum to the medioventral surface of the temporal lobe. Due to this shift in position, the fibers of the fornix must span a much larger distance to reach their terminal targets and this is reflected in the volume of the complex.

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